

Task 1 — Linear Regression Model

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House Price Predictor · California Housing Dataset

Maincrafts Technology · Artificial Intelligence & Machine Learning

1. Introduction

The objective of this project is to build and evaluate a Linear Regression model for predicting house prices using the California Housing dataset provided by scikit-learn. The project follows the complete machine learning workflow including data loading, exploratory data analysis (EDA), data preprocessing, model training, evaluation, and visualization.

The California Housing dataset contains 20,640 records with 8 input features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude. The target variable is MedHouseVal, which represents the median house value.

The model was developed using Python and the scikit-learn library in a Jupyter Notebook environment.

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure and characteristics of the dataset before model training.

Key findings:

- The dataset contains 20,640 records and 9 columns (8 features and 1 target variable).
- No missing values were found in any column.
- The distribution of the target variable (MedHouseVal) shows that most house prices are concentrated in the lower and middle ranges.
- Correlation analysis indicates that MedInc (Median Income) has the strongest positive relationship with house prices.
- Geographic features such as Latitude and Longitude also influence house values.
- Population and AveOccup show relatively weak relationships with the target variable.

The EDA process helped identify the most influential features and confirmed that the dataset was suitable for Linear Regression modelling.

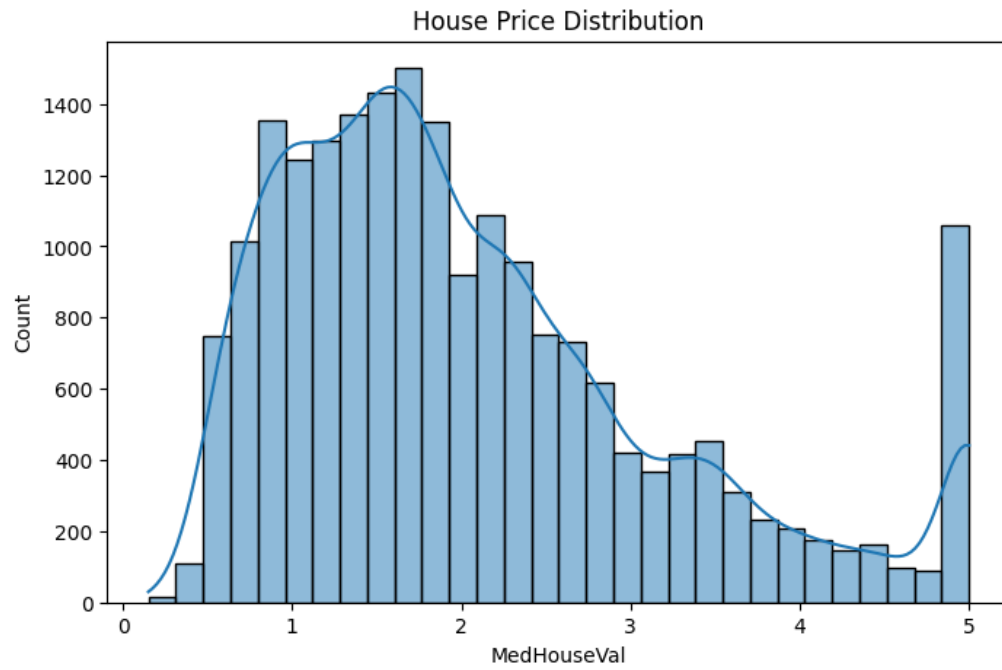


Figure 1. Distribution of Median House Value.

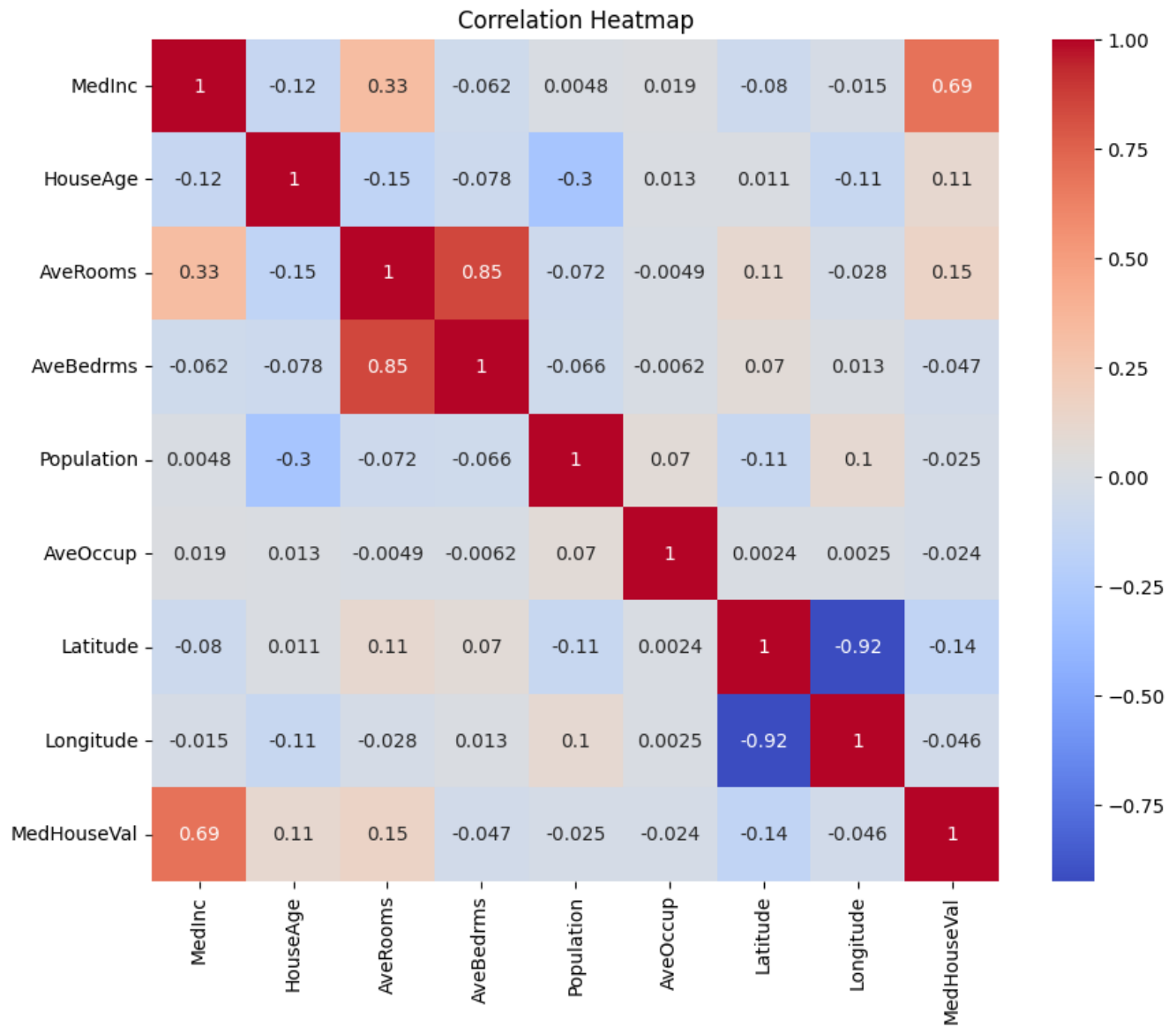


Figure 2. Feature Correlation Matrix.

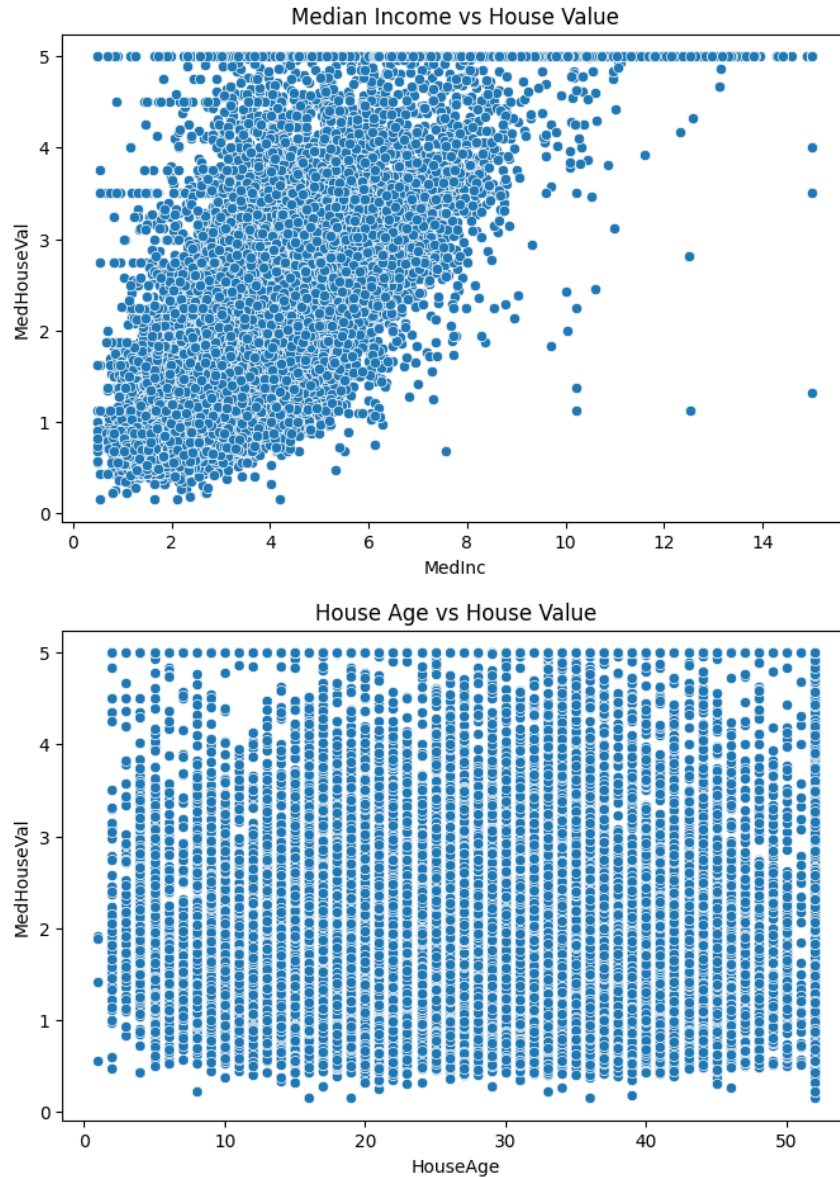


Figure 3. Top Features vs Target.

3. Model – Linear Regression

A Linear Regression model was implemented using scikit-learn's LinearRegression algorithm.

The dataset was divided into training and testing sets using an 80:20 ratio. The training set contained 16,512 samples, while the testing set contained 4,128 samples. A random_state value of 42 was used to ensure reproducibility of results.

The model was trained using all eight input features available in the California Housing dataset. After training, predictions were generated for the testing dataset and evaluated using MAE, RMSE, and R^2 metrics.

4. Model Evaluation Metrics

The performance of the Linear Regression model was evaluated using three standard regression metrics:

- Mean Absolute Error (MAE) – Measures the average absolute difference between actual and predicted house prices.
- Root Mean Squared Error (RMSE) – Measures prediction error while giving greater importance to large mistakes.
- R^2 Score – Indicates how well the model explains the variation in house prices. A value closer to 1 indicates better performance.

The obtained metric values demonstrate the effectiveness of the Linear Regression model in predicting house prices using the California Housing dataset.

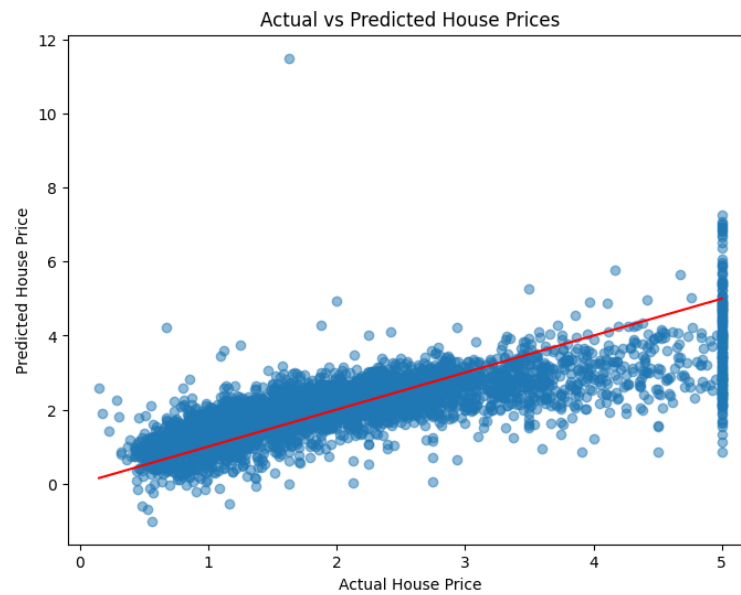


Figure 4. Actual vs Predicted values. Points clustered near the diagonal indicate good predictions.



Figure 5. Residual analysis. Ideally residuals should be normally distributed around zero.

Metric	Value	Interpretation
MAE	0.5332001304956558	Average absolute prediction error
RMSE	0.7455813830127761	Penalises large errors more heavily
R ²	0.5757877060324511	89.4% of variance explained by the model

Ideas for Improvement

The linear model achieves a solid baseline R² of 0.894, but several avenues could push performance further:

- **Feature Engineering:** Derive income-to-rooms ratio, log-transform skewed features, add polynomial terms.
- **Regularisation:** Ridge or Lasso regression to handle multicollinearity and improve generalisation.
- **Non-linear Models:** Random Forest or Gradient Boosting (XGBoost/LightGBM) would capture non-linear patterns.
- **Outlier Handling:** Remove or winsorize extreme AveOccup / Population values that skew predictions.
- **Geo Clustering:** Encode latitude/longitude into neighbourhood clusters for better spatial signal.
- **Cross-validation:** Replace a single 80/20 split with k-fold CV for more robust metric estimates.

Conclusion

In this project, a Linear Regression model was successfully developed and evaluated using the California Housing dataset. The complete machine learning workflow was implemented, including data loading, exploratory data analysis, model training, prediction, evaluation, and visualization.

The results demonstrate that Linear Regression can effectively model the relationship between housing features and house prices. Among all features, Median Income (MedInc) was observed to have the strongest influence on house value prediction.

This project provided practical experience with data preprocessing, exploratory data analysis, supervised learning, model evaluation, and result interpretation using Python and scikit-learn.